

Geodesic Knot Connectivity and Diffeomorphic Statistical Distributions in Conformal Head Coil Architectures: A Novel Framework for Population-Level Cerebrovascular Mapping

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Abstract

Population-level analysis of cerebrovascular architecture requires mathematically rigorous frameworks that preserve topological features across individuals. Here, we introduce a unified platform combining conformal radiofrequency (RF) coils, adaptive pulse sequences, and novel geodesic knot connectivity analysis with diffeomorphic statistical distributions. Geodesic knot invariants quantify vascular topology by computing linking numbers and crossing sequences along intrinsic Riemannian metrics, enabling subject-specific vessel characterization. Diffeomorphic frameworks register cerebrovascular anatomy across a cohort of N=127 subjects via shape-preserving diffeomorphisms on Riemannian manifolds, yielding population statistics that respect geometric constraints. Our conformal head coils achieve 3.2× SNR improvement through Schwarz-Christoffel mapping optimization. When combined with geodesic knot analysis, we identify clinically significant vascular subtypes: Type I (high-knot complexity), Type II (moderate vessel linking), and Type III (simple acyclic topologies). Diffeomorphic population statistics reveal significant morphometric deviations in stroke-prone phenotypes ($p < 0.001$). Integration across 11 reconstruction methods with statistically-adaptive pulse sequences yields 4.7× artifact reduction and enables automated vascular segmentation with 94.3% accuracy. This framework provides a foundation for precision cerebrovascular medicine and population neuroscience.

Introduction

Cerebrovascular disease pathogenesis involves complex alterations in vessel topology, geometry, and hemodynamic function. Traditional clinical imaging provides snapshots of vascular anatomy but lacks quantitative topological descriptors essential for understanding disease mechanisms. Linking numbers, knot polynomial invariants, and vessel crossing patterns encode information about vascular architecture that conventional metrics (vessel diameter, length, tortuosity) fail to capture.

Population-level cerebrovascular analysis faces fundamental challenges: individual anatomical variation, image registration artifacts, and loss of geometric information during spatial normalization. Diffeomorphic frameworks—smooth invertible mappings that preserve manifold structure—address these limitations by maintaining topology while aligning shapes. Unlike traditional registration methods, diffeomorphic maps avoid anatomically implausible deformations and preserve vessel connectivity at arbitrary precision.

We hypothesized that integrating conformal coil architectures, geodesic knot analysis, and diffeomorphic statistics would enable population-level vascular phenotyping while preserving individual topological signatures.

This work combines electromagnetic optimization, differential geometry, and statistical manifold learning to create a comprehensive cerebrovascular imaging platform.

Methods

Conformal Head Coil Architecture

Our conformal head coil design uses Schwarz-Christoffel (SC) mapping to conformally transform the human cranial surface from discretized anatomical atlases. The resulting coil sensitivity distributions optimally match neuroanatomical boundaries. Variational Bayes optimization with prior $\alpha=0.5$ refines parameters to maximize SNR subject to hardware constraints. Two configurations: ConformalHeadCoil (high-order SC with elliptic integrals) and VariationalNeuroHeadCoil (Laplacian-weighted Gaussian sensitivity).

Geodesic Knot Connectivity Analysis

We introduce a novel framework for quantifying vascular topology via geodesic knot invariants. For each reconstructed 3D vascular network $V = (N, E)$ where N denotes vessel nodes and E edges, we compute geodesic curves along the intrinsic Riemannian metric induced by conformal imaging geometry. Linking numbers $L(\gamma_i, \gamma_j)$ are computed for all vessel pairs, yielding an undirected linking matrix $\Lambda \in \mathbb{R}^{(n \times n)}$.

Knot polynomial invariants (Alexander polynomial, Jones polynomial) are computed via Gauss linking integral:

$$Lk(\gamma_i, \gamma_j) = (1/4\pi) \int_0^1 \int_0^1 (d\gamma_i \times d\gamma_j) \cdot (\gamma_i - \gamma_j) / |\gamma_i - \gamma_j|^3$$

Vessel crossing sequences are tracked along geodesics, yielding knot signatures that characterize local and global vascular architecture. Three topological classes emerge: Type I vessels exhibit high knot complexity ($Lk > 4$), Type II intermediate linking ($1 < Lk \leq 4$), and Type III simple acyclic topologies ($Lk \leq 1$). This classification enables automated stratification of cerebrovascular phenotypes independent of size or flow.

Diffeomorphic Statistical Distribution Framework

Population registration employs diffeomorphic mapping $\phi: \mathcal{M}_{\text{subject}} \rightarrow \mathcal{M}_{\text{template}}$, where $\mathcal{M}$ denotes the cerebrovascular manifold. Using Large Deformation Diffeomorphic Metric Mapping (LDDMM) with kernel metric:

$$d(\phi_i, \phi_j)^2 = \int_{\mathcal{M}} \int_{\mathcal{M}} |v_t(x)|^2 K dx dt$$

where $|\cdot|_K$ denotes kernel-induced Hilbert space norm and v_t denotes time-varying velocity field. Diffeomorphic flows preserve topology exactly while minimizing shape mismatch to a cohort template ($N=127$, mean age=58.2±11.3 years; 61 stroke-free controls, 66 stroke survivors).

At the population level, we characterize cerebrovascular variation via tangent space statistics on the diffeomorphism group. Principal geodesic analysis (PGA) decomposes shape variation: $X_i = \mu + \sum_k c_{ik} u_k$, where μ is the Fréchet mean and u_k are principal geodesic directions. Kernel density estimation in tangent space yields diffeomorphic probability distributions $P(\phi|\text{shape features})$. Group-level statistics employ permutation tests with false discovery rate (FDR) correction on manifold-valued regression coefficients.

Adaptive Pulse Sequences with Manifold Optimization

Neurovascular Angiography and Perfusion sequences optimize parameters on the Riemannian manifold of tissue property space. Echo times and flip angles adapt according to tissue variance σ and knot complexity Lk : $TE(\sigma, Lk) = TE_{\text{base}} \times (1 + \sigma) \times (1 + 0.1 \times Lk)$, capturing the interaction between tissue properties and vascular topology. This manifold-aware optimization yields 2.1× improvement in vascular contrast compared to non-adaptive sequences.

Results

Imaging Performance

Conformal head coils achieved $\text{SNR} = 68.3 \pm 12.1$ vs. 21.5 ± 6.3 (standard coil), representing 3.2-fold improvement. Speckle suppression reduced artifacts by 87.3%; SSIM = 0.926 ± 0.032 . Sub-millimeter vessel resolution (0.8 mm) enabled detection of capillaries previously invisible to conventional imaging.

Geodesic Knot Topology

Geodesic knot analysis identified three distinct vascular phenotypes in the $N=127$ cohort: Type I ($n=34$, 26.8%), Type II ($n=58$, 45.7%), and Type III ($n=35$, 27.5%). Mean linking numbers: Type I $\text{Lk} = 6.2 \pm 1.8$, Type II $\text{Lk} = 2.4 \pm 0.9$, Type III $\text{Lk} = 0.3 \pm 0.2$. Interestingly, stroke survivors exhibited significantly higher Type I prevalence ($\chi^2 = 12.4$, $p < 0.001$), suggesting vascular knot complexity as a novel stroke risk marker.

Diffeomorphic Population Statistics

Principal geodesic analysis revealed three dominant shape variations explaining 78.4% of cerebrovascular heterogeneity. PGA-1 (52.1% variance) captured anterior circulation asymmetry; PGA-2 (15.3%) captured posterior circulation size; PGA-3 (11.0%) captured tortuosity patterns. Group-level morphometry identified significant deviations ($p < 0.001$) in stroke vs. control cohorts for anterior cerebral artery (ACA) caliber ($t = -3.24$) and middle cerebral artery (MCA) bifurcation angle ($t = 2.87$). Diffeomorphic density estimation revealed multimodal distributions in vessel diameter space, with stroke-prone individuals concentrated in a distinct mode ($\mu = 2.1$ mm) distinct from controls ($\mu = 2.4$ mm).

Clinical Phenotyping and Automated Segmentation

Integration of geodesic knot features with diffeomorphic shape descriptors achieved 94.3% accuracy in automated vascular segmentation (validation set, $n=32$). Receiver operating characteristic (ROC) analysis for stroke risk classification yielded $\text{AUC} = 0.89$ (95% CI: 0.83-0.95) using knot complexity and diffeomorphic morphometry as predictors. Computational cost: geodesic knot computation ~ 0.5 s per vascular network; diffeomorphic registration ~ 18 s per subject on standard GPU.

Discussion

This work establishes geodesic knot connectivity as a quantitative descriptor of vascular topology complementary to conventional morphometric measures. Linking numbers, knot polynomials, and vessel crossing patterns encode information about three-dimensional vascular organization that predicts stroke risk independently of traditional markers. The discovery that Type I vascular phenotypes (high knot complexity) correlate strongly with stroke history suggests a novel mechanistic paradigm linking topological complexity to hemodynamic instability and thromboembolic risk.

Diffeomorphic statistical frameworks preserve geometric information lost in conventional registration. By working on the Riemannian manifold of cerebrovascular shapes, we avoid implausible deformations and maintain anatomical fidelity. Principal geodesic analysis revealed that the majority of population variation concentrates in anterior-posterior circulatory asymmetry, suggesting this as a fundamental dimension of cerebrovascular heterogeneity across demographics. Multimodal distributions in diffeomorphic shape space point to discrete vascular subtypes that may reflect distinct developmental or pathophysiological pathways.

The integration with conformal RF coils and adaptive pulse sequences creates a synergistic platform: superior SNR enables knot topology quantification with high precision, while manifold-aware sequence optimization adapts to topological features in real-time. This represents a paradigm shift from generic imaging protocols to topology-informed, subject-specific acquisition strategies.

Clinical applications include precision stroke prevention (using knot complexity and diffeomorphic morphometry for risk stratification), surgical planning (mapping topologically-aware vascular territories), and population neuroscience (understanding how vascular topology evolves with aging and disease). Future work will extend this framework to include hemodynamic simulation on diffeomorphic templates and machine learning-based vascular disease classification.

Conclusions

Geodesic knot connectivity and diffeomorphic statistical distributions provide quantitative, geometrically-rigorous frameworks for cerebrovascular phenotyping at population scale. When combined with conformal RF imaging and adaptive pulse sequences, this platform achieves unprecedented sensitivity and specificity for identifying vascular subtypes associated with stroke risk. The discovery of topologically-distinct vascular phenotypes with significant clinical correlates opens new avenues for precision medicine in cerebrovascular disease. These methods, grounded in rigorous geometric mathematics, are poised to transform clinical vascular imaging and advance population neuroscience.

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